

MANGALORE UNIVERSITY
COMPUTER SCIENCE TEACHERS' ASSOCIATION
(MUCSTA)

MANGALORE UNIVERSITY

**Choice Based Credit System Semester Scheme with
Multiple Entry and Exit Options in the UG
Programmes under NEP 2020**

Bachelor of Computer Applications (BCA)
2021-22 Onwards

I Semester BCA

Question Bank

Fundamentals of Computers [2-7]
Programming in C [8-11]
Mathematical Foundation [12-22]
Office Automation [23-27]

FUNDAMENTALS OF COMPUTERS

UNIT I

Questions carrying 2 Marks

1. What is meant by generations of Computers?
2. Expand UNIVAC, ENIAC, EDVAC, EDSAC, BIOS, EPROM (1 Mark Each)
3. Define Artificial Intelligence. List different technologies related to AI.
4. What is an Expert System?
5. What is parallel processing ?
6. What is a Microcomputer?
7. What is a Supercomputer?
8. What are Registers?
9. What is a Bus?
10. What is System Bus? Name its functional groups
11. What is the main memory? Name two types.
12. Differential RAM and ROM
13. What is Cache memory?
14. What are pointing devices? Give example.
15. What is Trackball? Why it is used?
16. What is Joystick? Mention its use.
17. What are output devices? List different forms of output devices.
18. What is a hard copy? What is a soft copy?
19. Write different purposes of the light pen.
20. What is a plotter? List different types of plotters.
21. Define resolution and refresh rate of monitors

3-6 marks questions

1. List and explain characteristics of Computers. (6)
2. List and explain in brief the characteristics of First-Generation Computers. (6)
3. List and explain in brief the characteristics of Second-Generation Computers. (6)
4. List and explain in brief the characteristics of Third-Generation Computers. (6)
5. List and explain in brief the characteristics of Fourth-Generation Computers. (6)
6. List and explain in brief the characteristics of Fifth-Generation Computers. (6)
7. Write a note on
 - a. Mainframes
 - b. Supercomputers.(6)
8. Draw the block diagram of a Computer System and explain the components of a Computer System. (6)
9. List and explain the applications of Computers in different fields (6)
10. Write a note on
 - a. Keyboard (3)
 - b. Pointing devices (6)
 - c. Touch Screen (3)
 - d. Trackpad (3)
 - e. Digital Camera (3)
 - f. Scanners (4)
 - g. Dot Matrix Printers (3)
 - h. Ink-jet Printers (3)
 - i. Monitors (6)
 - j. Laser Printer (3)
 - k. Plotters (3)

11. Differentiate LCD and CRT monitors (4)
12. Explain System Bus and its functional groups. (6)

UNIT II

Questions carrying 2 Marks

1. Define Software.
2. Define System Software. Give example.
3. Define Application Software. Give example.
4. Define Utility Software? List any two System Utility functions.
5. Differentiate compilers and interpreters.
6. What is Machine Language?
7. What is Assembly Language?
8. What is Loader?
9. List basic tasks of a loader.
10. Define Algorithm
11. Define flowcharts
12. List any 4 symbols used in flow charts along with their purpose
13. What is pseudocode?
14. List any two limitations of pseudocode.
15. What is Word Processor? Give example for Word Processors
16. What is Spreadsheet Software? Give example for Spreadsheet Software
17. What are image editors? Give example
18. What is Database Management System? Give example for Database Management System
19. What is Presentation Software? Give example for Presentation Software
20. What is Desktop Publishing? Give example for Desktop Publishing Software

4-6 Marks Questions

1. Define Operating System. List and explain basic functions of an Operating System (6)
2. Write a note on
 - a. Device Drivers
 - b. Language Translators (6)
3. What is System Utility? Explain any five functions of System Utility (6)
4. What is Application Software? Explain any two commonly used application software. (6)
5. List and explain functions of System Utilities. (6)
6. Write a note on
 - a. Word Processors (3)
 - b. Image Editors (3)
 - c. Presentation Software (3)
 - d. Desktop Publishing Software (3)
7. What are assemblers? List and explain different functions of Assemblers. (6)
8. Briefly explain different phases of the Program Development Cycle (6)
9. Discuss flowcharts. List and explain the benefits of flowcharts (6)
10. Write an algorithm to display maximum of three numbers (4)
11. Write an algorithm to calculate the factorial of a number (4)
12. Discuss flowcharts. List and explain the limitations of flowcharts (6)
13. Discuss pseudocode. List and explain the advantages and limitations of pseudocode (6)
14. Explain the purpose of Pseudocode. List and explain keywords commonly used in pseudocode (6)

15. List and explain different benefits and limitations of pseudocode. (6)
16. List and explain features of a good programming language (6)
17. List and explain in brief the advantages and limitations of Machine Languages (6)
18. List and explain in brief the advantages and limitations of Assembly Languages (6)
19. List and explain the advantages and limitations of High-level Languages (6)
20. Explain different 4GLs. Also, mention the advantages and disadvantages of 4GLs (6)

UNIT III

Questions carrying 2 Marks

1. Convert the following Octal Numbers to Binary and Hexadecimal:
 - a. 154, 273, 256, 362, 267, 654, 745, 631, 537, 465 (2 Marks each)
2. Convert the following Hexadecimal Numbers to Binary and Octal:
 - a. (AB2)₁₆, (98D)₁₆, (AFD)₁₆, (1BC)₁₆, (AE5)₁₆ (2 Marks each)
3. Convert the following Binary Numbers to Octal and Hexadecimal:
 - a. 1010100₍₂₎, 101011₍₂₎, 1011110₍₂₎, 1111110₍₂₎, 110110010₍₂₎ (2 Marks each)
4. Write the BCD and Excess-3 Codes for the following Decimal Numbers
 - a. 74, 57, 38, 43, 83 (2 Marks each)
5. Write BCD and 2421 code of the following
 - a. 34514, 98 76, 254, 87 (2 Marks each)
6. BCD and 84-2-1 code of the following
 - a. 98, 74, 57, 63, 38 (2 Marks each)
7. Write 2421 and 84-2-1 code of the following
 - a. 78, 94, 83, 65, 91 (2 Marks each)
8. Write 9's and 10's Complement of the following numbers
 - a. 5243, 8126, 874, 8790, 4567, 4562.89, 128.536 (2 Marks each)
9. Write 1's and 2's Complement of the following numbers
 - a. (1110101)₂, (101001)₂, (11001)₂, (111010)₂, (10101011)₂, (101011.110)₂ (2 Marks each)
10. Perform the following operations:
 - a. (100101.11)₂ + (11011.101)₂
 - b. (1011101.11)₂ - (11011.101)₂
 - c. (1000011)₂ + (1101101)₂
 - d. (101110111)₂ - (11011101)₂
 - e. (1101101)₂ - (11011)₂
 - f. (11101)₂ + (11011)₂ (2 Marks each)
11. State any two postulates of Boolean algebra
12. Prove that
 - a. $X + X = X$
 - b. $X \cdot X = X$ (2 Marks each)
13. Write the dual of
 - a. $x + x = x$
 - b. $x + x' = 1$ (1 Mark each)

Questions Carrying 3-6 Marks

1. Perform the following subtractions using 9's and 10's Complement Methods
 - a. 841 - 329 b. 325 - 641 c. 821 - 413 d. 8056 - 3250 e. 6327 - 8659
 - f. 6408 - 543 g. 272 - 354 (4 Marks each)
2. Perform the following subtractions using 1's and 2's Complement Methods (4 Marks each)
 - a. (1001)₂ - (1011)₂
 - b. (10011)₂ - (1001)₂

- c. $(10110)_2 - (11101)_2$
 d. $(110101)_2 - (100101)_2$
3. Perform following conversions.
 (4 Marks each)
- a. $(BCA.8)_{16} = ()_{10}$
 b. $(201)_8 = ()_2$
 c. $(451)_{10} = ()_8$
 d. $(152.A1)_{16} = ()_{10}$
 e. $(896.5)_{10} = ()_{16}$
- e. $(101011)_2 - (111001)_2$
 f. $(1011.001)_2 - (110.10)_2$
- f. $(345)_8 = ()_2$
 g. $(915)_{10} = ()_8$
 h. $(244.5)_{10} = ()_2$
 i. $(111011010.11)_2 = ()_{10}$
 j. $(625.5)_{10} = ()_{16}$
4. Using Venn diagram prove that
- a. $X = XY + X$.
 b. $x(y+z) = xy + xz$
 c. $A + (B + C) = (A + B) + C$ (4 Marks each)
5. Write a note on parity codes with example. (4)
 6. Write a note on alphanumeric codes. (4)
 7. Prove state and prove commutative law using truth table. (4)
 8. State and prove distributive law. (AND) over + (OR) using truth table. (6)
 9. State and prove distributive law +(OR) over. (AND) using truth table. (6)
 10. State and prove DeMorgan's theorems for two variables. (6)
 11. State and prove DeMorgan's theorems for three variables. (6)
 12. State and prove any two theorems of Boolean algebra. (6)
 13. Write the procedure to perform 1's complement subtraction with example (6)
 14. Write the procedure to perform 2's complement subtraction with example (6)
 15. Write the procedure to perform 9's complement subtraction with example (6)
 16. Write the procedure to perform 10's complement subtraction with example (6)

UNIT IV

Questions Carrying 2 Marks

- Write the Truth Table and Logic Symbol of the following
 AND gate, OR gate, NOT gate, NAND gate, NOR gate, XOR gate, XNOR gate
 (2 Marks each)
- Write the Truth Table and Logic expression of the following
 AND gate, OR gate, NOT gate, NAND gate, NOR gate, XOR gate, XNOR gate
 (2 Marks each)
- Draw the logic circuit for $F(X, Y, Z) = XY + X'Z'$ using basic gate.
- Draw the logic circuit for $F(X, Y, Z) = (X + Y)'$ using basic gate.
- Define Sum term and Product term.
- What are minterms and maxterms?
- What do you mean by the duality of the Boolean algebra? Give an example.
- What do you mean by duality principle? Write the dual of $F = (x+y)(x+y')$
- Write the general structure of 2 variables K-Map.
- Write the general structure of 3 variables K-Map
- Write the general structure of 4 variables K-Map
- Define Pair, QUAD, and OCTET in K-Map.
- Draw the logic circuit for the following Boolean functions using basic gates
 - $F(X, Y, Z) = XY + X'Z'$

- b. $F(X, Y, Z) = (X+Y)'$
 - c. $F(A, B, C) = A'B + BC$
 - d. $F(X, Y, Z) = (X' + Y)Z$
 - e. $F(A, B, C) = AB' + A'C$ (2 Marks each)
14. Find the complement of Boolean expression $F = (X+Y)(X+Y'+Z)$
 15. Find the complement of $F1 = x'yz' + x'y'z$
 16. Find the complement of $F2 = x(y'z' + yz)$
 17. Prove $x + x'y = x + y$
 18. Simplify $xy + xy'$
 19. What do you mean by standard form? Give an example.
 20. What do you mean by canonical form? Give an example.

Questions Carrying 3-6 or more Marks

1. Express the Boolean function $F = XY' + X'Z$ as sum of minterms and product of maxterms. (6)
2. Express the Boolean function $F = WXY' + X'Y'Z$ as sum of minterms and product of maxterms. (6)
3. Using theorems and Postulates simplify $F(A, B, C, D) = A'B(D' + C'D) + B(A + A'CD)$ (4)
4. Using Boolean Theorems and postulates, prove the following: (4 marks each)
 - a. $x + x'y = x + y$
 - b. $x'y'z + x'yz + xy' = x'z + xy'$
5. Using truth table, prove the following: (4 marks each)
 - a. $x + x'y = x + y$
 - b. $x'y'z + x'yz + xy' = x'z + xy'$
6. Minimize the following Boolean expression using Boolean postulates and Theorems $F(A, B, C) = (A+B)(A+C)$ (4)
7. Implement Boolean function $F = x'y'z + x'yz + yz'$ with basic gates and also write the truth table. (6)
8. Obtain the Truth Table of Boolean function and draw the logic diagram (6)
 - a. $F = xy + xy' + y'z$
 - b. $F = xy + x'y' + y'z$ (6)
9. Express the Boolean function, $F(X, Y, Z) = XY + X'Z$ as sum of minterms and product of maxterms (6)
10. Implement the following Boolean function using gates and also write the truth table.
 - a. $F = xyz'$
 - b. $F = xy' + x'z$ (4 marks each)
11. Express the Boolean function $F(A, B, C) = A + B'C$ as sum of minterms and product of maxterms (6)
12. Minimize the Boolean expression $F(x, y, z) = x'y'z + x'yz + xy'$ and draw the logic diagram for the minimized expression (6)
13. Solve the following using K-Maps and draw the logic diagram for minimized expression. Also draw the logic diagram for the simplified Boolean expression. (6 marks each)
 - a. $F(A, B, C) = \sum(1, 3, 6, 7)$
 - b. $F(A, B, C) = \sum(0, 1, 2, 4, 5, 6)$
 - c. $F(A, B, C) = \sum(0, 2, 4, 6)$
 - d. $F(A, B, C) = \sum(1, 2, 5, 7) + \sum(0, 4, 6)$
 - e. $F(A, B, C) = \sum(0, 1, 6, 7) + \sum(3, 4, 5)$
 - f. $F(P, Q, R, S) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$
 - g. $F(A, B, C, D) = \sum(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

- h. $F(A, B, C, D) = \Sigma(1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d(0, 2, 14)$
- i. $F(A, B, C, D) = \Sigma(0, 2, 8, 10, 14) + \Sigma d(5, 15)$
- j. $F(A, B, C, D) = \Sigma(3, 4, 5, 7, 9, 13, 14, 15)$
- k. $F(W, X, Y, Z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$
- l. $F(A, B, C, D) = \Sigma(0, 3, 4, 7, 8) + \Sigma d(10, 11, 12, 13, 14, 15)$
- m. $F(w, x, y, z) = \Sigma(4, 5, 6, 8, 9, 12, 13, 14)$
- n. $F(w, x, y, z) = \Sigma(8, 9, 10, 11, 12, 13, 14)$
- o. $F(A, B, C, D) = \Sigma(0, 1, 4, 8, 9, 10)$
- p. $F(A, B, C, D) = \Sigma(0, 1, 3, 4, 5, 7, 12, 13, 15)$

14. What are universal gates? Prove that NAND is universal gate. (6)

15. What are universal gates? Prove that NAND is universal gate. (6)

PROGRAMMING IN C

UNIT – I

2 marks questions

- 1) List any two important features of C
- 2) What are tokens? Give any two examples.
- 3) Give the structure of C program.
- 4) List steps involved in executing a C program
- 5) What are keywords? List any four keywords.
- 6) What are identifiers? Give example for a valid and invalid identifier name.
- 7) Differentiate keywords and identifiers.
- 8) List the rules to be followed for naming an identifier/variable.
- 9) What are constants? Give examples for integer constants.
- 10) Differentiate constants and variables.
- 11) Differentiate integer and real constants. Give example
- 12) Differentiate character and string constants.
- 13) List and specify the meaning of any four backslash character constants.
- 14) Provide the syntax for declaring a variable. Give an example.
- 15) What is initialization? Give an example.
- 16) What do you mean by symbolic constants? How to define them?
- 17) Provide the syntax of printf(). Give an example.
- 18) Provide the syntax of scanf(). Give an example.
- 19) List any four unformatted I/O functions available in C
- 20) List with use any four format codes used with scanf()/printf()
- 21) Specify the value of x after evaluating following expression
 $x=3+4-7*8/5\%10;$
- 22) Specify the value of x after evaluating following expression
float x =1.5;
y=3
 $x=y/2+y*8/y-y+x/3;$
- 23) Specify the value of x after evaluating following expression
int y=3, z=4, x;
float t=4.2;
 $x=z*z/y+y/2*t+2+t;$

6 marks questions

- 1) List and explain features of C [6].
- 2) Give Structure of C program and Explain each section. [6].
- 3) Explain structure of C program with programming example [6]
- 4) Write a note on Tokens in 'C' [6].
- 5) Write a note on Constants in 'C' [6].
- 6) Explain in brief classification of a) integer types and b) floating point types
- 7) List and explain primary data types available in 'C' [6].
- 8) Explain with examples declaring, initializing and assigning value to variable [6].
- 9) Explain with example #define statement in 'C'. List rules apply to it. [6].
- 10) Explain with example reading, writing a character and strings in C [6]
- 11) Explain with example formatted Input in C [6].
- 12) Explain with example formatted Output in C [6]
- 13) Explain printf() function with its syntax and example [6]
- 14) Explain scanf() function with its syntax and example [6]
- 15) Write a program to read two real values and perform addition, subtraction, multiplication and division operation on them [6].

UNIT – II

2 marks questions

- 1) What are operators? Specify their use.
- 2) List different categories of operators available in C
- 3) What is integer arithmetic? Give an example.
- 4) What is real arithmetic? Give an example
- 5) What is relational expression? Give an example
- 6) What is logical expression? Give an example
- 7) Differentiate pre/post increment and decrement operators
- 8) List short-hand assignment operators available in C
- 9) What is the difference between = and == operators?
- 10) Give the syntax of conditional operator. Give an example.
- 11) List special operators available in C along with their meaning
- 12) Define arithmetic expression. Give an example.
- 13) Specify the precedence of arithmetic operators in C
- 14) What do you mean by type conversion? Mention its types.
- 15) List any four built-in Mathematical functions available in C
- 16) Write the syntax of simple if statement. Give an example.
- 17) Give the syntax of goto statement. Give an example
- 18) Define the term looping. What are the different looping statements available in C?
- 19) Differentiate entry-controlled and exit-controlled loops
- 20) Differentiate break and continue statements

6 marks questions

- 1) List and explain arithmetic, logical and relational operators available in C (6)
- 2) List and explain a) conditional operator, b) short-hand assignment and c) increment & decrement operators available in C (6)
- 3) Write a note on bitwise operators available in C (6)
- 4) Explain with example evaluation of arithmetic expression in C (6)
- 5) Explain precedence of arithmetic operators with the help of an example expression (6)
- 6) Explain the concept of type conversion in C, Give examples (6)
- 7) Explain in brief implicit and explicit type conversion with example (6).
- 8) Explain different forms of if statement (6)
- 9) Explain nested if statement with the help of a program example (6)
- 10) Explain else..if ladder with the help of a program example (6)
- 11) Explain switch statement with syntax and a program example [6]
- 12) Explain while statement with syntax and example (6)
- 13) Explain do..while statement with syntax and example (6)
- 14) Explain for statement with syntax and example (6)
- 15) Explain the following: (6)
 - i. Nested loops
 - ii. Infinite loops

UNIT – III

2 marks questions

- 1) What is an array? List its types.
- 2) List any four advantages of arrays.
- 3) Differentiate one-dimensional and two-dimensional arrays
- 4) Write a statement that declares an integer array of 10 element size and initializes it with values.
- 5) How to initialize a one-dimensional array in C. Give an example.
- 6) How to initialize a two-dimensional array in C. Give an example.
- 7) What is a string? How to declare a variable to hold string value in C?

- 8) What is the difference between reading string value using scanf(“%s”) and gets() functions?
- 9) List any four-string handling functions available in C
- 10) List any four-character handling functions.
- 11) How to declare and initialize string variable?
- 12) What are the possible values that the function strcmp() can return? What do they mean?
- 13) How to initialize a string array in C. Give an example.
- 14) Mention suitable string functions to do the following:
 - a. To find length of the string
 - b. To copy one string to another string
 - c. To add two strings
 - d. To reverse given string
- 15) What is pointer? Specify its use.
- 16) List any four advantages of pointers
- 17) Specify how to declare a pointer variable and initialization of it?
- 18) List any four rules to be followed when performing operations on pointer variable
- 19) What do you mean by scale factor in pointer increment?

6 marks questions

- 1) Explain one-dimensional array with the help of suitable code example [6].
- 2) Explain two-dimensional array with the help of suitable code example [6].
- 3) Explain the different ways available for the initialization of string variable [6].
- 4) Write a program to sort elements of an integer array [6].
- 5) Write a program to search for an element in an integer array [6].
- 6) Explain with example how to declare, initialize and use strings in C [6].
- 7) List and explain any four-string handling functions available in C [6]
- 8) List and explain any four-character handling functions available in C [6]
- 9) What is a pointer variable? How is it declared, initialized and used in a program?
- 10) Discuss the rules applied when performing operations on pointer variable [6]
- 11) Illustrate the use of pointers with arrays [6]
- 12) What do you mean by pointer arithmetic? Explain how it is differed from other integer arithmetic [6]
- 13) Explain how pointers are used to access string? [6]
- 14) Write a note on pointers increments and scale factor[6]

UNIT – IV

2 marks questions

- 1) List any two advantages of using functions?
- 2) Give the syntax used for declaring a function in C
- 3) Give the syntax used for defining a function in C
- 4) What are the actual and formal arguments?
- 5) How function returns value ? Give example.
- 6) Differentiate function declaration and function definition.
- 7) List different categories of functions available in C
- 8) Give syntax and example of calling a function C.
- 9) What do you mean by nesting of functions? Give example.
- 10) What is a recursive function?
- 11) Write recursive function to calculate factorial of a given number.
- 12) What is structure? Why it is required?
- 13) Give the general format used for defining a structure
- 14) Give the general format used for defining a union
- 15) How to Access individual members of structure? Give example.

- 16) What is union? How it is declared?
- 17) Differentiate structure and Union.

6 marks questions

- 1) Explain with example how to declare, define and call a function in C [6]
- 2) Explain with the help of a code example declaring, defining and calling a function in C [6]
- 3) Explain with example function with no arguments and no return values [6]
- 4) Explain with example function with arguments and no return value [6]
- 5) Explain with example function with no arguments but a return value [6]
- 6) Explain with example functions with arguments and a return value [6]
- 7) Write a note on a) nesting of functions b) recursive functions [6]
- 8) Explain with example defining a structure, declaring structure variable and accessing structure members [6].
- 9) Explain with the help of code example array of structures [6]
- 10) Explain with code example array within structure [6]
- 11) Write a note on nesting of structures [6]
- 12) Explain with example defining a union, declaring union variable and accessing union members [6]

MATHEMATICAL FOUNDATION

Unit -I

Questions carrying two mark each

1. Define logarithm.
2. State any two laws of logarithms.
3. Change into logarithmic form (one mark each)
a. $6^{-1} = \frac{1}{6}$ b. $2^4 = 16$ c. $\sqrt[3]{8} = 2$ d. $a^0 = 1$
4. Change into exponential form (One mark each)
a. $\log_4 64 = 3$ b. $\log_5 \frac{1}{625} = -4$ c. $\log_{\sqrt{2}} 16 = 8$ d. $\log_{\sqrt{3}} 9 = 4$
5. Prove that $\log \frac{x}{y} + \log \frac{y}{z} + \log \frac{z}{x} = 0$
6. Simplify $\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 18$
7. Find the value of x if
a. $\log_5 x = 3$ b. $\log_a x = 0$
8. Find the value of a, if $\log_a \sqrt{2} = \frac{1}{6}$
9. Find the value of (i) $\log_4 256$ (ii) $\log_2 64$
10. Simplify: $\log_a b \cdot \log_b c \cdot \log_c d \cdot \log_d a$
11. Prove that $\log_y x \log_z y \log_x z = 1$
12. If $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find the value of $\log 6$?
13. If $\log 2 = 0.3010$ find the value of $\log_8 25$?
14. Write Binomial theorem.
15. Find the middle term in the expansion of $\left(\frac{a}{x} - bx\right)^{12}$
16. Write distance formula.
17. Write the coordinates of any point lying on
a. X axis b. Y axis.
18. Find the distance between (Each question carries 2 marks)
a. $(5,3), (1,-1)$ d. $(5,7), (2,11)$
b. $(-4,-2), (3,-5)$ e. $(9,-1), (-2,10)$
c. $(5,5), (-3,3)$ f. $(-1,4), (5,-4)$
19. Find the coordinates of the mid-point of the join of the points $(-2, 5)$ and $(6, 3)$.
20. Find the co-ordinates of the centroid of the triangle whose vertices are $(3,2), (-1,-4)$ and $(-5,6)$
21. Find the coordinates of the centroid whose vertices are $(2,3)$ $(-5,2)$ and $(1,7)$.
22. Find the coordinates of the centroid whose vertices are $(3,-2)$ $(-1,-4)$ and $(-5,6)$.
23. Write the formula to calculate the area of a triangle.
24. Find the coordinates of the point which divides internally the line joining $(6, -5)$ and $(-7, -15)$ in the ratio 4:7.
25. Find the coordinates of the point which divides externally the line joining $(-7, 1)$ and $(3, 6)$ in the ratio 4:3

26. Find the coordinates of the point which divides internally the line joining (6, -5) and (-7, -15) in the ratio 4:3.
27. Find the equation of the straight line passing through the origin and making with the x-axis an angle of 45° .
28. Find the equation of the straight line passing through the origin and making with the x-axis an angle of 60° .
29. Find the equation of the circle whose centre is (4,5) and radius is 7.
30. Find the centre and radius of the circle whose equation is $x^2+y^2-4x+8y-5=0$.
31. Find the centre and radius of the circle whose equation is $x^2+y^2-x+\frac{2}{3}y+\frac{1}{3}=0$.

Questions carrying 4 or 6 marks

1. Solve for x if $\log_x 3 + \log_x 9 + \log_x 729 = 9$. (4)
2. Prove that $\frac{2}{3} \log 8 - \frac{1}{2} \log \frac{1}{4} - 3 \log 2 = 0$ (6)
3. Prove that $\frac{\log \sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5} = \frac{3}{2}$ (6)
4. Solve without using log table show that $\log \frac{81}{8} - 2 \log \frac{3}{2} + 3 \log \frac{2}{3} + \log \frac{3}{4} = 0$ (6)
5. Simplify $\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 18$ (6)
6. Evaluate $\log \frac{41}{35} + \log 70 - \log \frac{41}{2} + 2 \log 5$ (6)
7. Find the value of $\frac{2 \log 6 + 6 \log 2}{4 \log 2 + \log 27 - \log 9}$ (6)
8. Prove that $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80} + \log \frac{1}{2} = 0$ (6)
9. If $\log_2 x + \log_4 x + \log_{16} x = \frac{21}{4}$, find x. (4)
10. If $x = \log_7 27$ $y = \log_5 7$ $z = \log_3 5$ prove that $x.y.z=3$ (4)
11. Prove that $\log_y(x^3) \log_z(y^3) \log_x(z^3) = 1$ (4)
12. Expand $\left(x - \frac{1}{x}\right)^5$ using Binomial theorem. (4)
13. Expand $(3x - y)^4$ using Binomial theorem. (4)
14. Expand $\left(\frac{x}{3} + \frac{2}{y}\right)^4$ using Binomial theorem. (4)
15. Find the 11th term in the expansion of $(y+4x)^{30}$. (4)
16. Find the fifth term in the expansion of $\left(\frac{3x}{4} + \frac{4}{3x}\right)^{12}$. (4)
17. Find the 7th term in the expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^9$. (4)
18. Find the middle term/terms in the expansion of $\left(3x - \frac{2x^2}{3}\right)^7$ (6)
19. Find the 7th term in the expansion of $\left(x^3 - \frac{1}{x}\right)^9$. (4)
20. Find the middle term in the expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^8$. (4)
21. Find the middle term in the expansion of $\left(x - \frac{1}{2y}\right)^{10}$. (4)
22. Find the middle term in the expansion of $\left(x - \frac{1}{x}\right)^{19}$. (6)
23. Find the middle term in the expansion of $\left(\frac{a}{x} + bx\right)^{12}$. (4)

24. Find the middle term in the expansion of $\left(2x - \frac{4}{x}\right)^7$. (6)
25. Prove that the points (6, 6) (2, 3) and (4, 7) form a right angled triangle. (4)
26. Prove that the points (0, 3) (-2,1) and (-1,4) form a right angled triangle. (4)
27. Show that the points (-1, 2) (3, -1) and (2, 6) form an isosceles triangle. (4)
28. Show that the points (4, 3) (7, -1) and (9, 3) form an isosceles triangle. (4)
29. Show that the points (1, -1) (-1, 1) and $(-\sqrt{3}, -\sqrt{3})$ are the vertices of an equilateral triangle. (4)
30. Show that the quadrilateral with points (2, -1) (3, 4) (-2, 3) and (-3, -2) is a rhombus. (6)
31. Show that the points (0,-5) (3,0) (0,5) and (-3,0) is a rhombus. (6)
32. Show that (8, 3) (2, -1) (0, 1) and (6, 5) are the vertices of a parallelogram. (6)
33. Show that (-2, -1) (1, 0) (4, 3) and (1, 2) are the vertices of a parallelogram. (6)
34. Show that (4, -5) (8, 1) (14, -3) and (10, -9) is a square. (6)
35. Show that the points (4, -5) (8, 1) (14, -3) and (10, -9) are the vertices of a square. (6)
36. Show that (-1, -1) (2, -1) (2, 2) and (-1, 2) is a square. (6)
37. Find the coordinates of a point which divides the line joining two points P (8, 9) and Q (-7, 4) internally in the ratio 2:3 and externally in the ratio 4:3. (4)
38. Find the coordinates of the point which divides the line joining (5, 2) and (7, 9) internally in the ratio 2:7 and externally in the ratio 1:6. (4)
39. Find the coordinates of the point which divides internally and externally the line joining (2, -4) and (7, 1) in the ratio 2:3. (4)
40. Find the area of the triangle represented by (2, 3), (5, 7) and (-3, 4). (4)
41. Find the area of the triangle represented by (0, 0), (1, 2) and (-1, 2). (4)
42. Find the area of the triangle represented by (2, -1), (-3, -4) and (0, 2). (4)
43. The coordinates of two points A and B are (-1, 2) and (2, -1) respectively. Find the equation and the slope of the line AB. (4)
44. Find the equation and slope of line passing through (2, 4) and (4, -1). (4)
45. Find the equation and slope of line passing through (5, -7) and (2, 3). (4)
46. Find the equation of straight lines passing through (-5,-7) parallel to x axis and parallel to y axis. (4)
47. Find the equation of straight lines passing through (-8, 5) parallel to x axis and parallel to y axis. (4)
48. Find the equation of straight line passing through the point (-3, 1) perpendicular to the line $5x-2y+7=0$. (4)
49. Find the equation of straight line passing through (2, 4) and parallel and perpendicular to $5x-7y=100$. (6)
50. Find the equation of straight line : (6)
 - (i) parallel to $2x - 3y - 5 = 0$ and passing through (4,5)
 - (ii) perpendicular to $2x + 3y + 4 = 0$ and passing through (3,-2)

51. Find the equation of straight line passing through the point (4, 2) parallel to the line $5x - 7y + 11 = 0$. (4)
52. Find the equation of straight line passing through the point (-2, 7) parallel to the line $x - 5y + 3 = 0$. (4)
53. Find the equation of straight line passing through the point (-1, 2) perpendicular to the line $4x - 3y + 7 = 0$. (4)
54. Find the equation of straight line passing through the point (-2, 7) parallel to the line $x - 5y + 3 = 0$. (4)
55. Find the equation of straight line passing through the point (4, 5) parallel to the line $2x - 3y - 5 = 0$. (4)
56. Find the equation of straight line passing through the point (3, -2) perpendicular to the line $2x + 3y + 4 = 0$. (4)
57. Find the equation of tangent and normal to the circle $x^2 + y^2 - 6x + 8y + 17 = 0$ at (1, -2). (4)
58. Find the equation of tangent and normal to the circle $x^2 + y^2 + 8x + 10y - 105 = 0$ at (1, 6). (4)

Unit II

Questions carrying two mark each

1. Define angle and radian.
2. Represent the following angles in radians (1 mark each)
 - a. 90° b. 45° c. 60° d. 135° e. 225° f. 120°
 - g. 270° h. 40° i. $112^\circ 30'$
3. Write the reciprocal of $\sin\theta$ and $\cos\theta$
4. Write the reciprocal of $\operatorname{cosec}\theta$ and $\tan\theta$
5. Write the reciprocal of $\sec\theta$ and $\cot\theta$
6. Find the value of $\sin 90^\circ + \cos 90^\circ + \tan 45^\circ$
7. Find the value of $\sin^2 90^\circ + \cos^2 90^\circ + \tan 45^\circ$
8. Define the limit of a function and write the value of $\lim_{x \rightarrow 0} \frac{x^n - a^n}{x - a}$
9. Find the limit of $\frac{4x^4 + 3x^2 - 1}{x^3 + 7}$ when $x \rightarrow 1$
10. Define continuity of a function.
11. Prove that the function $x^2 + 4x - 2$ is continuous at $x = 1$.
12. If $y = 2x + x^2$ then find $\frac{dy}{dx}$?
13. Differentiate (i). $\frac{1}{\sqrt{x}}$ (ii). $\frac{-3}{\sqrt[3]{x^7}}$ with respect to x
14. Differentiate $7x^4 + 3x^3 - 9x + 5$ with respect to x .
15. Differentiate $x + \frac{4}{x} - \frac{2}{x^7}$ with respect to x .
16. Write the value of $\int e^x dx$ and $\int x^n dx$
17. Write the value of $\int e^x dx$ and $\int \frac{1}{x} dx$
18. Integrate i. $x^{\frac{6}{5}}$ ii. $\sqrt[3]{x^4}$

19. Find $\int 5x^2 dx$
20. Find $\int (3 - 2x - x^4)dx$
21. Find $\int (x^2 - 1)^2 dx$
22. Integrate $\sqrt{x} - \frac{1}{\sqrt{x}}$
23. Integrate $7x^2 - 3x + 8$
24. Find $\int_{-1}^1 (2x^2 - x^3)dx = ?$
25. Find $\int_6^{10} \left(\frac{dx}{(x+2)}\right)$
26. Prove that $\int_4^8 \frac{dx}{x-3} = \log 5$

Questions carrying 4 or 6 marks.

1. Express both in degrees and radians the angles of a triangle whose angles are to each other as 1:2:3. (4)
2. Express both in degrees and radians the angles of a triangle whose angles are to each other as 2:3:5. (4)
3. Find the angles of a triangle in radian, if the angles are in the ratio 1:2:6 (4)
4. If $\sin\theta = \frac{15}{17}$ and θ is an acute angle find the values of other trigonometric functions. (6)
5. If $\cos\theta = \frac{24}{25}$ and θ is an acute angle find the values of other trigonometric functions. (6)
6. If $\cot\theta = \frac{24}{7}$ find the values of other trigonometric functions. (6)
7. If $\sin\theta = \frac{15}{17}$ find the values of other trigonometric functions. (6)
8. Find the value of other trigonometric functions of the specified angle, if $\sin\theta = \frac{7}{25}$ (6)
9. If θ is in 4th quadrant and $\cos\theta = \frac{5}{13}$ find the value of $\frac{13\sin\theta + 5\sec\theta}{5\tan\theta + 6\csc\theta}$ (4)
10. If $\sin\theta = \frac{8}{17}$ find $\tan\theta + \sec\theta$ (4)
11. If $\tan\theta = \frac{4}{5}$ find the value of $\frac{2\sin\theta + 3\cos\theta}{4\cos\theta + 3\sin\theta}$ (4)
12. If $\sec\theta = \frac{13}{5}$ and θ is in the 4th quadrant, find the value of $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$ (4)
13. If $\sin\theta = \frac{8}{17}$ find $\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}$ (4)
14. If $\sin\theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the value of $\frac{\sec\theta - \tan\theta}{\csc\theta + \cot\theta}$ (4)
15. If $\sec\theta = \sqrt{2}$ and $\frac{3\pi}{2} < \theta < 2\pi$, find the value of $\frac{1 + \tan\theta + \csc\theta}{1 + \cot\theta - \csc\theta}$ (4)
16. Find the values of $\frac{\tan 45^\circ}{\csc 30^\circ} + \frac{\sec 60^\circ}{\cot 45^\circ} + \frac{5}{2} \cdot \frac{\sin 90^\circ}{\cos 0^\circ}$ (4)
17. Find the values of $\cos^2 0^\circ + \cos^2 30^\circ + \cos^2 45^\circ + \cos^2 60^\circ + \cos^2 90^\circ$ (6)
18. Find the values of $\frac{1}{3}\sin^2 60^\circ - \frac{1}{2}\sec 60^\circ \tan^2 30^\circ + \frac{4}{3}\sin^2 45^\circ \tan^2 60^\circ$ (6)
19. Prove that $4(\sin^4 30^\circ + \cos^4 60^\circ) - 3(\cos^2 45^\circ - \sin^2 90^\circ) - 2 = 0$ (6)
20. If $\tan^2 45^\circ - \cos^2 60^\circ = x \sin 45^\circ \tan 60^\circ$, find x. (6)
21. Find x from the equation, $x \sin 30^\circ \cos^2 45^\circ = \frac{\cot^2 30^\circ \sec 60^\circ \tan 45^\circ}{\csc^2 45^\circ \csc 30^\circ}$ (6)
22. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 + 2x - 5}{x^2 - 9}$ (4)
23. Evaluate $\lim_{x \rightarrow 0} \left(\frac{4x^4 + 5x^3 + 7x^2 + 6x}{5x^5 + 7x^2 + x} \right)$ (6)
24. Evaluate $\lim_{x \rightarrow 2} \frac{2x^2 - 7x + 6}{5x^2 - 11x + 2}$ (4)
25. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 5x + 6}$ (4)

26. Evaluate $\lim_{x \rightarrow 2} \frac{2x^2 - 5x + 2}{x^2 - 3x + 2}$ (4)
27. Let $f(x) = \frac{x^2 - 4}{x - 2}$ for $x < 2$ (4)
 $f(x) = 4$, for $x = 2$
 $f(x) = 2$, for $x > 2$, discuss the continuity at $x = 2$.
28. Let $f(x) = \frac{x^2 - 9}{x - 3}$ Prove that f is discontinuous at $x = 3$. (6)
29. Let $f(x) = \frac{x^2 - 4}{x - 2}$ for $x < 2$ (4)
 $f(x) = 2$, for $x \geq 2$, discuss the continuity at $x = 2$.
30. Let $f(x) = \frac{x^2 - 4}{x - 2}$ for $0 \leq x < 2$ (6)
 $f(x) = 2$, for $x = 2$,
 $f(x) = x + 1$, When $x > 2$, discuss the continuity at $x = 2$.
31. Let $f(x) = \frac{x^2 - 4}{x - 2}$ for $0 \leq x < 2$ (6)
 $f(x) = 2$, for $x = 2$
 $f(x) = \frac{3x + 2}{x}$, When $x > 2$, discuss the continuity at $x = 2$.
32. Differentiate with respect to x (i) $4x^3 - 7x^4$ (ii) $\frac{4}{3}x^3 - \frac{6}{7}x^7 + 4x^{-3}$
33. Differentiate $9x^4 - 7x^3 + 8x^2 - \frac{8}{x} + \frac{10}{x^3}$ with respect to x .
34. Differentiate $(3x^2 + 1)(x^3 + 2x)$ with respect to x . (6)
35. Differentiate $\frac{x^2 + 4}{3x - 7}$ with respect to x . (6)
36. Differentiate $\frac{x^2 - 1}{x^2 + 1}$ with respect to x . (6)
37. Differentiate $\frac{3x^2 + 5x}{7x + 4}$ with respect to x . (6)
38. Differentiate $\frac{(x + 1)(2x - 1)}{(x - 3)}$ with respect to x . (6)
39. Evaluate $\int (4x^3 + 3x^2 - 2x + 5) dx$ (4)
40. Evaluate $\int \left(\sqrt{x} - \frac{1}{2}x + \frac{2}{\sqrt{x}} \right) dx$ (4)
41. Integrate $7x^2 - 3x + 8 - \frac{1}{\sqrt{x}} + \frac{1}{x} + \frac{1}{x^2}$ (6)
42. Find the value of $\int_2^4 (3x - 2)2 dx$ (4)
43. Find the value of $\int_{-3}^{-1} \left(\frac{1}{x^2} - \frac{1}{x^3} \right) dx$ (4)
44. Find the value of $\int_2^4 (9x^2 - 12x + 4) dx$ (4)

Unit III

Questions carrying two mark each

1. Define matrix with one example.
2. Define square matrix with an example.
3. Define row matrix with an example
4. Define column matrix with an example
5. Define diagonal matrix with an example
6. Define unit matrix with an example
7. Define null matrix with an example

8. Define triangular matrix with an example
9. Define sub matrix with an example
10. Define scalar matrix with an example
11. Define symmetric matrix with an example
12. Define skew-symmetric matrix with an example
13. Define transpose of a matrix with an example.
14. Prove that, matrix $A = \begin{pmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 7 \end{pmatrix}$ is symmetric.
15. If matrix $A = \begin{bmatrix} 4 & -3 \\ 8 & -2 \\ -1 & 0 \end{bmatrix}$, find $4A$.
16. Define equality of matrices with an example.
17. Find x and y if $\begin{pmatrix} x+y & 2 \\ 1 & x-y \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 7 \end{pmatrix}$
18. Find $(x \ y)$ if $(4 \ 5) + (x \ y) = (7 \ 3)$
19. Find $(x \ y)$ if $(1 \ -9) - (2 \ -3) = (x \ y)$
20. Find $(x \ y)$ if $(x \ y) - (0 \ -1) = (5 \ 4)$
21. Classify the following matrices:
 - (i) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
 - (ii) $\begin{pmatrix} 3 & 0 & 0 \\ 1 & -4 & 0 \\ 9 & 5 & 10 \end{pmatrix}$
 - (iii) $\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$
 - (iv) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
22. If $A = \begin{bmatrix} 2 & 3 & -1 \\ -3 & -1 & 4 \end{bmatrix}$ and $B = \begin{pmatrix} 2 & -1 & 3 \\ 7 & 2 & -1 \end{pmatrix}$, find $A+B$
23. If $A = \begin{bmatrix} 2 & 3 & -1 \\ -3 & -1 & 4 \end{bmatrix}$ and $B = \begin{pmatrix} 2 & -1 & 3 \\ 7 & 2 & -1 \end{pmatrix}$, find $A-B$
24. If $A = (1 \ 2 \ 3 \ 4)$ and $B = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$ find AB .
25. Define orthogonal matrix.
26. Let a matrix $A = \begin{pmatrix} 6 & 5 \\ 3 & 2 \end{pmatrix}$, find $|A|$
27. Let a matrix $A = \begin{pmatrix} 5 & 5 \\ 3 & 2 \end{pmatrix}$, find $|A|$
28. Let a matrix $A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$, find $|A|$
29. Let $A = \begin{pmatrix} 3 & 4 & 7 \\ -2 & 5 & 6 \\ 8 & -1 & 9 \end{pmatrix}$, find the minor of the element 5.
30. Let $A = \begin{pmatrix} 3 & 4 & 7 \\ -2 & 5 & 6 \\ 8 & -1 & 9 \end{pmatrix}$, find the minor of the element 3.
31. Define co-factor of the element.

32. If $A = \begin{pmatrix} 3 & 4 & 7 \\ -2 & 5 & 6 \\ 7 & 3 & -9 \end{pmatrix}$, find the co-factors of elements 6 and -9.

33. Define rank of a matrix with an example.

34. Find rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$

35. Find rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 0 & 5 \end{bmatrix}$

36. Define echelon form of a matrix with an example

37. Define normal form of a matrix with an example

38. Define equivalence of matrices.

Questions Carrying 4 or 6 marks

1. If $A = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 6 & 3 \\ 1 & 4 & 5 \end{pmatrix}$, then find the value of $2A + 3B$. (4)

2. If $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ -3 & 2 \end{pmatrix}$, find AB and BA . Is $AB = BA$? (4)

3. Given the matrices A, B, C , $A = \begin{pmatrix} 2 & 3 & -1 \\ 3 & 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, $C = (1 \quad -2)$, Verify that

$$(AB)C = A(BC). \quad (6)$$

4. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{pmatrix}$ Show that $A^3 - 3A^2 - A + 9I = O$ (6)

5. Given $A = \begin{pmatrix} 2 & 0 & 4 \\ 6 & 2 & 8 \\ 2 & 4 & 6 \end{pmatrix}$, $B = \begin{pmatrix} 8 & 4 & -2 \\ 0 & -2 & 0 \\ 2 & 2 & 6 \end{pmatrix}$, $C = \begin{pmatrix} 8 & 2 & 0 \\ 0 & 2 & -6 \\ -8 & 4 & -10 \end{pmatrix}$, Compute the following:

a) $A + (B + C)$ b) $A - B - C$ c) $3A + 2B - 3C$ d) $3B + 2A$ (any one) (4)

6. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 4 & 9 \end{pmatrix}$, Show that $AB = O$ (4)

7. If $A = \begin{pmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ -3 & -3 & 3 \end{pmatrix}$ show that AB and CA are null matrices. (6)

8. If $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, show that $A^2 = 2A$ (4)

9. If $A = \begin{pmatrix} 2 & -3 & 1 \\ 4 & 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -2 & 4 \\ 1 & 3 & -5 \end{pmatrix}$, Show that $(A + B)' = A' + B'$ (6)

10. For each of the following matrices verify that $(A')' = A$ (each 4)

a) $\begin{pmatrix} 2 & 8 & 4 \\ 8 & 6 & -1 \\ 4 & -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 9 & 2 & -7 & 2 \\ 7 & 8 & 5 & 6 & 0 \end{pmatrix}$

11. If $A = \begin{pmatrix} 3 & -3 & 0 \\ 6 & 3 & 9 \\ 12 & 3 & 24 \end{pmatrix}$, $B = \begin{bmatrix} 2 & 3 & 0 \\ 6 & -9 & 3 \\ 3 & 3 & -3 \end{bmatrix}$, verify that $(A + B)' = A' + B'$ (6)

12. If $A = \begin{pmatrix} 3 & -3 & 0 \\ 6 & 3 & 9 \\ 12 & 3 & 24 \end{pmatrix}$, $B = \begin{bmatrix} 2 & 3 & 0 \\ 6 & -9 & 3 \\ 3 & 3 & -3 \end{bmatrix}$, verify that $(AB)' = B'A'$ (6)

13. If $A = \begin{bmatrix} 8 & 16 & -4 \\ -4 & 0 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 12 & 16 & 20 \\ -4 & 8 & 28 \\ 8 & 4 & 0 \end{bmatrix}$ Compute $(AB)'$ and $B'A'$ (6)
14. Show that the matrix $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ is orthogonal. (6)
15. Find the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ (6)
16. Find the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ and verify the theorem
- $A (\text{Adj } A) = (\text{Adj } A) A = |A| I_3$ (8)
17. Find the adjoint of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & -5 \end{pmatrix}$, verify $A (\text{Adj } A) = (\text{Adj } A) A = |A| I_2$ (8)
18. Find the adjoint of the matrix $\begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 6 \\ 0 & 1 & -3 \end{bmatrix}$ and verify the theorem
- $A (\text{Adj } A) = (\text{Adj } A) A = |A| I_3$ (8)
19. Find the adjoint of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$ and verify the theorem
- $A (\text{Adj } A) = (\text{Adj } A) A = |A| I_3$ (8)
20. If $A = \begin{pmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$, show that $\text{Adj } A = 3A'$ (8)
21. Find rank of the matrix $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 5 \\ 2 & 4 & 8 \\ 0 & 2 & 3 \end{bmatrix}$ (6)
22. Find rank of the matrix $C = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 4 & 6 \\ 0 & 6 & 9 \end{bmatrix}$ (6)

Unit 4

Questions carrying 2 marks

- Define inverse of a matrix.
- Define characteristic equation of a matrix. Find the characteristic equation of a matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$
- State Cayley Hamilton Theorem. Write its mathematical definition.
- Define method of reduction.
- What does condition say, system is consistent in Cramer's rule.
- Define arithmetic progression. Give one example.
- Write formula for finding n^{th} term and sum to n terms in arithmetic progression.
- Define arithmetic mean. Write the formula to find AM.
- The sum of a certain number of terms in AP is 5500. The first and last terms are 100 and 1000. Find the number of terms.
- Define geometric progression. Give one example.

11. Write formula for finding n^{th} term and geometric mean between 'a' and 'b' in geometric progression.
12. Write formula for finding sum to n terms in geometric progression.
13. If the value of a machine is Rs 1,60,000 and it depreciates at the rate of 10% per year what will be its value at the end of 5 years?
14. Find n^{th} term and the sum to n terms of the series: 4, 12, 36, 108, ...
15. Find the 40th term of an AP whose 9th term is 465 and 20th term is 388.
16. Find the sums of all integers between 200 and 500 which are divisible by 7.
17. Find the sum of all natural numbers between 200 and 500 which are divisible by 17.
18. The 10th term of an AP 184 and 16th term is 160. Find the 45th term.
19. The n^{th} term of a series is $7n-3$. Show that the series is an AP.
20. The n^{th} term of a series is $7n-3$. Find the first term and the common difference.

Questions carrying 4 or more marks

1. Compute the inverse of the matrix: $\begin{bmatrix} 1 & 0 & -4 \\ -2 & 2 & 5 \\ 3 & -1 & 2 \end{bmatrix}$ (6)
2. Let $B = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$ and $C = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ Find X if $BX = C$ (6)
3. If $A = \begin{pmatrix} -1 & -5 \\ -2 & 3 \end{pmatrix}$, then show that $A^{-1} = -\frac{1}{13} \begin{pmatrix} 3 & 5 \\ 2 & -1 \end{pmatrix}$ (6)
4. If $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$, Find matrix B if AB equals $\begin{pmatrix} 22 & 6 \\ 11 & 3 \end{pmatrix}$ (6)
5. If $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$, Find matrix B if AB equals $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (6)
6. If $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$, Find matrix B if AB equals $\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$ (6)
7. Show that the matrix $\begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ satisfies its characteristic equation (8)
8. Show that the matrix $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ satisfies its characteristic equation (8)
9. Show that the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ satisfies its characteristic equation (8)
10. Show that the matrix $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ satisfies its characteristic equation (8)
11. Show that the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ satisfies its characteristic equation (8)
12. Using matrices solve the equations, (8)

$$\begin{aligned} x + y + 2z &= 4 \\ 2x - y + 3z &= 9 \\ 3x - y - z &= 2 \end{aligned}$$
13. Solve by matrix method (8)

$$2x + 4y + z = 5; \quad x + y + z = 6; \quad 2x + 3y + z = 6$$

14. Solve the following equations by matrix method: (8)

$$2x - 3y = 3 \text{ and } 4x - y = 11$$

15. Using matrices, solve the equations: (8)

$$x + y + z = 6, \quad x + 2y + 3z = 14, \quad -x + y - z = -2$$

16. Using matrices, solve the equations: (8)

$$x + y + 2z = 4, \quad 2x - y + 3z = 9, \quad 3x - y - z = 2$$

17. Using matrices, solve the equations: (8)

$$2x + 4y + z = 5, \quad x + y + z = 6, \quad 2x + 3y + z = 2$$

18. Solve the following equations by using method of reduction: (8)

$$x + y + z = 6; \quad x - y + z = 2; \quad 2x + y - z = 1$$

19. Solve the equations using method of reduction: (8)

$$x + y + z = 6; \quad x + 2y + 3z = 14; \quad x + 4y + 9z = 36$$

20. Solve the equations using method of reduction: (8)

$$2x + 4y - z = 9; \quad 3x + y + 2z = 7; \quad x + 3y - 3z = 4$$

21. Using Cramer's rule, solve the following equation (8)

$$x + y - 4 = 0, \quad 2x - 3y - 8 = 0$$

22. Using Cramer's rule, solve the following equation (6)

$$x + y + z = 6, \quad x - y + z = 2, \quad 3x + 2y - 4z = -5$$

23. Solve $x + 2y + 3z = 6$, $2x + 4y + z = 17$, $3x + 2y + 9z = 2$ using Cramer's rule. (6)

24. Using Cramer's rule, solve the following equation (6)

$$x + y + z = 6, \quad 2x + y - z = 1, \quad x + y - 2z = -3$$

25. Using Cramer's rule, solve the following equation (6)

$$x + y + z = 6, \quad x - y + z = 2, \quad 3x + 2y - 9z = -5$$

26. Using Cramer's rule, solve the following equation (6)

$$x + y + z = 11, \quad 2x - 6y - z = 0, \quad 3x + 4y + 2z = 0$$

27. The rate of monthly salary of a person increases annually in AP. It is known that he was drawing Rs 200 a month during the 11th year of service and Rs 380 during the 29th year. Find his starting salary and the rate of annual increments. (4)

28. Find the three numbers in AP whose sum is 12 and the sum of whose cubes are 408. (6)

29. Find the three numbers in AP whose sum is 24 and whose product is 440. (6)

30. Find the three numbers in AP whose sum is 30 and whose product is 840. (6)

31. A firm produces 1000 sets of TV during its first year. The total sets produced at the end of 5 years are 7000. Estimate the annual rate of increase in production, if the increase in each year is uniform. (4)

32. I propose to take 30 consecutive terms of the series $100+99+98+97+\dots$. At what term must I begin so that the sum of the series is 1155? (4)

33. Find the 3 numbers in GP whose sum is 21 and whose product is 216. (6)

34. Find the sum to n terms of the geometric series $3, 2, \frac{4}{3}, \frac{8}{9}, \dots$ (4)

35. Three numbers whose sum is 18 are in AP. If 2,4,11 are added to them respectively, the resulting numbers are in GP. Determine the numbers. (6)
36. The sequence of three numbers a, b, c is an AP whose sum is 18. If a and b are each increased by 4 and c is increased by 36, the new numbers form a GP. Find a, b , and c . (6)
37. A man borrows Rs 5,115 to be paid in 10 monthly instalments. If each instalment is double the value of the last, find the value of the first and last instalments. (4)
38. Find three numbers in GP whose product is 1000 and sum is 35. (6)
39. The n^{th} term of a series is $2(1.5)^{n-1}$. Show that the series is a GP. (4)
40. Find three numbers in GP whose sum is 14 and product is 64. (6)

OFFICE AUTOMATION

Unit-1

Two Mark Questions:

1. Define Computer Software?
2. What are the components of the computer?
3. Differentiate System Software and Application Software.
4. What do you mean by software piracy? List the types of software piracy?
5. List any four software terminologies.
6. Why MS Paint program is used?
7. Why Control panel is used?
8. Define Antivirus.
9. Mention different file management tools.
10. Define Office Button.
11. Define Firmware. Give an example.
12. Define Operating System.
13. Name any four Operating Systems.
14. What is File Explorer?
15. Why Compression tools used. Give an example.
16. What is MS Word?
17. What are the various options in MS Office User Interface?
18. Give the steps to correct Errors in MS Word
19. List out different Text Alignment Options.
20. Write a short note on Bold, Italic and Underline commands in MS Word

Long Answer questions

1. Briefly explain the software installation and uninstallation? (04)
2. Briefly explain software piracy? (04)
3. Briefly explain the windows desktop environment? (06)
4. Briefly explain Software Terminologies. (04)
5. Explain the various tools available in MS Paint toolbox? (06)
6. Briefly explain the Windows Explorer and File Explorer (04)
7. Explain MS Office user interface (06)
8. Briefly explain options in Office Button. (05)
9. Explain the concept Quick Access Toolbar, Page Layout Tab and Ruler. (06)
10. How to insert text and save document in MS Word. (04)
11. Explain How to select, copy, paste and delete text in MS Word. (04)
12. Briefly explain How to Correct Errors in MS Word. (04)
13. Explain how to change Font Style and Format Font Color in MS Word. (04)
14. Briefly explain different text cases and font alignments in MS Word. (04)
15. Briefly explain how to apply and customize style in MS Word. (04)
16. Briefly explain how to change page orientation and page size in MS Word. (04)
17. Briefly explain how to change page margin and insert page break in MS Word. (04)
18. Briefly explain various steps to insert a header and footer in MS Word document. (04)
19. Briefly explain how to edit and delete header and footer in MS Word. (04)
20. Explain insert table in MS Word. (04)

Unit-2

Two Mark Questions

1. What is MS Excel?
2. What is Workbook?
3. What is Worksheet?
4. What is formula bar?
5. What is the usage of status bar in the MS-Excel?
6. Write a note on word wrap in MS-Excel?
7. Define Ribbon in MS Excel.
8. Mention various types mathematical calculations.
9. What is function? List any three built in functions in MS-Excel
10. Mention any four DATE & TIME functions.
11. Mention any four TEXT functions.
12. List any four header and footer elements.
13. What is Header and Footer button in MS Excel?
14. Define Pivot table.
15. Mention different cell alignment options.
16. How to save and close excel file?
17. How to create borders in MS Excel?
18. What do you mean by sorting in MS-Excel?
19. How to add an image in MS-Excel?
20. How to add a clip Art in MS-Excel?
21. How to create charts in MS Excel?

Long Answer questions

1. Explain the Environment of MS-Excel? (06)
2. Explain the various methods to move around Worksheet. (04)
3. Briefly explain the concept Select Cells. (04)
4. Explain the various ways to edit a cell in MS-Excel. (04)
5. Briefly explain the various ways of Insert and Delete Columns and Rows in MS-Excel. (04)
6. Briefly explain merge and center in MS Excel. (04)
7. Briefly explain how to change font, font size and font color. (04)
8. Briefly explain the various formatting of text. (04)
9. Explain the concept of changing column width in MS-Excel. (04)
10. Explain the various formatting numbers in the MS-Excel. (04)
11. Briefly explain the mathematical calculation in MS-Excel. (06)
12. Briefly explain any four built in functions in MS-Excel. (04)
13. Briefly explain how to fill cell automatically and copy cells. (04)
14. Explain the step by step procedure to create the Chart In MS-Excel with example. (06)
15. Briefly explain headers and footers in MS Excel. (04)
16. Briefly explain the pivot table creation in MS-Excel. (06)
17. Briefly explain how to adding clip art and image in MS Excel. (04)
18. Explain the concept of printing the worksheet in MS-Excel. (04)

Unit-3

Two Mark Questions

1. Define Internet.
2. List any four internet terms.
3. Define internet relay chat.
4. List any two internet applications?

5. Differentiate web page and web site.
6. What is browser? Mention it's types.
7. What is Download and Upload?
8. List any four Internet Applications.
9. Define chat room.
10. What is Internet Telephony? Give an example.
11. Give any four features of Web browser.
12. Define E-mail.
13. Why back and forward buttons are used in Internet explorer?
14. What is MS PowerPoint?
15. What is usage of slides, placeholders and notes in MS PowerPoint?
16. What are the various options in view button?
17. Define Transition in MS PowerPoint.
18. What is Multimedia Presentation?
19. How to insert images into PowerPoint?
20. Define Animation in MS PowerPoint.

Long answer questions

1. Explain the features of the browser. (04)
2. Explain basic internet terms. (06)
3. Explain internet applications. (06)
4. Briefly explain the video conferencing. (04)
5. Briefly explain internet Explorer environment. (06)
6. What is instant messaging? Explain any three features of messenger. (04)
7. Write the structure of E- mail Address. (04)
8. Explain the advantage and disadvantages of E-mail. (04)
9. Explain MS PowerPoint Environment. (06)
10. Explain the concept view button with its types. (04)
11. Explain the steps to create PowerPoint Presentation. (06)
12. Explain how to work with Animation in MS PowerPoint. (05)
13. How to add Transition in MS PowerPoint? Explain. (05)
14. Explain the concept Slide sorter view and Print in MS PowerPoint. (04)
15. Explain the steps to insert images in MS PowerPoint. (04)
16. Explain steps to create an Outline. (04)

Unit-4

Two Mark Questions

1. Define Database?
2. What is DBMS (Data Base Management System)?
3. What is RDBMS (Relational Database Management System)?
4. Differentiate Field and Record.
5. What is Datatype? List out different types of data.
6. Define MS Access.
7. Define Database Window.
8. What is Database objects?
9. Define Macros.
10. Why Queries are used in Database?
11. Define Form in MS Access.
12. What are the different ways to modifying forms in MS Access?
13. Define Module in MS Access.

14. Define tables in MS Access.
15. Mention any four unique parts in MS Access environment.
16. What is the usage of field properties in Database?
17. Why Input Mask property is used?
18. Differentiate format property and Field property.
19. Define Primary key.

Long answer questions

1. Briefly explain the basic terms of the Database. (04)
2. Briefly explain introduction to MS Access. (04)
3. Write the steps to Creating and Saving a Database in MS-Access. (04)
4. Write the steps to open and closing a database in MS-Access. (04)
5. Briefly explain database objects. (04)
6. Explain MS Access environment. (06)
7. Write the steps to create the Table in MS-Access. (06)
8. Briefly explain data type and property. (04)
9. Explain steps to create adding and deleting records in a MS Access Table. (06)
10. With an example explain how to add primary key to a table. (04)
11. Explain steps to create query in MS-Access. (06)
12. Briefly explain the steps to create form using form-wizard in MS-Access (06)
13. Briefly explain the Macros in MS-Access. (06)
14. Briefly explain the reports in MS-Access. (06)
15. Using Query concept create a table named "STUDENT" with the fields Roll Number, Name, Gender, DOB, Address, Contact Number, Class, Sub_1 Mark, Sub_2 Mark, Sub_3 Mark, Total and add Primary Key (Roll No) then insert any five students data using INSERT query. (06)
